coupling $W_\ell U = GV$, we get

$$\begin{aligned}
\left(\mathcal{W}_2(\mu(W_\ell H_\ell), \mu(G/\sqrt{n}))\right)^2 &= \inf_{\text{all the couplings}} \mathbb{E}\|W_\ell U \mathrm{diag}(\sigma) V^\top - GVV^\top/\sqrt{n}\|_F^2 \\
&\leq \mathbb{E}\|GV\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)V^\top\|_F^2 \\
&= \mathbb{E}\mathrm{Tr}(GV\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)V^\top V\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)V^\top G^\top) \\
&= \mathrm{Tr}(V\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)V^\top \mathbb{E}\left[G^\top G\right]) \\
&= \mathbb{E}\mathrm{Tr}(\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)V^\top V) \\
&= \mathbb{E}\|\left(\mathrm{diag}(\sigma) - I/\sqrt{n}\right)\|_F^2 \\
&= \mathbb{E}\left[\sum_{i=1}^n (\sigma_i - 1/\sqrt{n})^2\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n \left(\sigma_i^2 - 1/n\right)^2 / \left(\sigma_i + 1/\sqrt{n}\right)^2\right] \\
&\leq n\mathbb{E}\left[\sum_{i=1}^n \left(\sigma_i^2 - 1/n\right)^2\right] \\
&\leq n\mathbb{E}\left[V^2(H_\ell)\right] \\
&\leq 2n\mathbb{E}\left[V(H_\ell)\right].
\end{aligned}$$

Appendix E. Orthogonality gap for the iterative initialization

Recall the proposed initialization for weights based on SVD decomposition $H_\ell = U_\ell \Sigma_\ell V_\ell^\top$.

$$W_\ell = \frac{1}{\|\Sigma_\ell^{1/2}\|_F} V_\ell' \Sigma_\ell^{-1/2} U_\ell^\top.$$

Here, we show that

$$V(H_\ell) > V(W_\ell H_\ell) \tag{17}$$

holds as long as $V(H_\ell) \neq 0$ and $\mathcal{A}_1(\alpha, \ell)$ holds. Given singular values H_ℓ, we get

$$\begin{aligned}
V(H_\ell) &= \sum_{i=1}^n \left(\sigma_i^2 - \frac{1}{n}\right)^2 \\
&= \sum_i \sigma_i^4 - 1/n,
\end{aligned} \tag{18}$$

where we used $\sum_{i=1}^n \sigma_i^2 = 1$ (see Eq. (2)). Now, we compute $V(H_\ell)$ using the singular values.

$$W_\ell H_\ell = \frac{1}{\|\Sigma_\ell^{1/2}\|_F} V_\ell' \Sigma_\ell^{1/2} V_\ell$$

Hence, the following holds

$$\begin{aligned}
V(W_\ell H_\ell) &= \sum_{i=1}^n \left(\frac{\sigma_i}{\sum_j \sigma_j} - \frac{1}{n}\right)^2 \\
&= \frac{1}{(\sum_{i=1}^n \sigma_i)^2} - 1/n
\end{aligned}$$

Combining with Eq. (18), we get

$$V(H_\ell) - V(W_\ell H_\ell) = \sum_{i=1}^{n} \sigma_i^4 - \frac{1}{(\sum_{i=1}^{n} \sigma_i)^2}.$$

To show that the right side of the above equation is positive, we need to prove

$$\sum_i \sigma_i^4 \left(\sum_{i=1}^{n} \sigma_i \right)^2 > 1 = \left(\sum_i \sigma_i^2 \right)^4$$

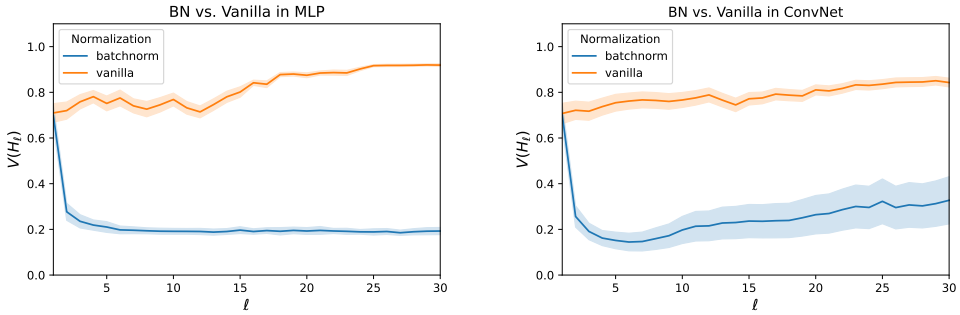
holds. Using Cauchy-Schwarz inequality, we get

$$\begin{aligned} \left(\sum_i \sigma_i^2 \right)^4 &= \left(\sum_i \sigma_i^{3/2} \sigma_i^{1/2} \right)^4 \\ &\leq \left(\sum_i \sigma_i \right)^2 \left(\sum_i \sigma_i^2 \sigma_i \right)^2 \\ &\leq \left(\sum_i \sigma_i \right)^2 \left(\sum_i \sigma_i^4 \right)^2, \end{aligned}$$

where the equality in the above inequality is met only for $\sigma_i^2 = \frac{1}{n}$ (so $V(H_\ell) = 0$) under $\mathcal{A}_1$.

Appendix F. Activation functions and the orthogonality

We have shown hidden representations become increasingly orthogonal through layers of BN MLPs with linear activations. We observed a similar results for MLP with ReLU activations and also a simple convolution network (see Figure 5). Here, we elaborate more on the role activation functions in our analysis.



(a) MLP (b) ConvNet

Figure 5. Orthogonality gap of layers for BatchNorm vs vanilla network with ReLU, for MLP and a basic convolutional network on CIFAR10. Batch size was set to 8, the number of hidden layers for both networks was set to 30, with 100 width for MLP and 100 channels for ConvNet in all hidden layers. For the ConvNet, the kernel size was set to 3, across all hidden layers, with zero-padding to make convolutional feature maps equal sized. The standard deviations were computed for 50 different randomly sampled batches.

For an arbitrary activation function F, the hidden representations make the following Markov chain:

$$Q_{\ell+1} = BN(W_\ell F(Q_\ell)).$$

F.1. **Conjecture.** Suppose that $\{Q_\ell\}$ is *ergodict* (Eberle, 2009), and admits a unique invariant distribution denoted by ν. We define function $L : \mathbb{R}^{d\times n} \to \mathbb{R}_+$ as

$$L(Q_\ell) = \left\| Q_\ell^\top Q_\ell - \mathbb{E}_{Q\sim\nu}\left[Q^\top Q\right]\right\|_F. \tag{19}$$

We conjecture that there exits an integer k such that for $\ell > k$ the following holds:

$$\mathbb{E}\left[L(Q_\ell)\right] = \mathcal{O}\left(\frac{1}{\sqrt{d}}\right). \tag{20}$$

Figure 6 experimentally validates that the above result holds for standard activations such as ReLU, sigmoid, and tanh.

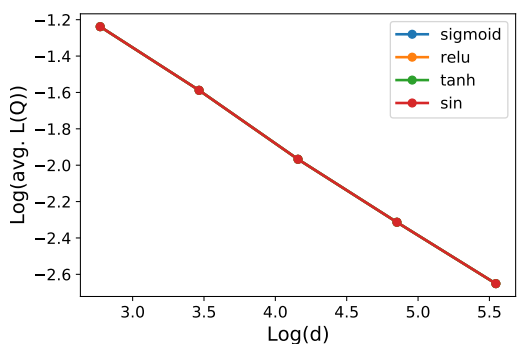


FIGURE 6. **Validating the conjecture.** Horizontal axis: $\log(d)$; vertical axis: $\log(\frac{1}{1000}\sum_{k=1}^{1000} L(Q_k))$. The difference between plots for the activation functions is negligible, so not visible.

F.2. **Odd activation.** We observe that Theorem 1 holds for odd activations such as sin and tanh. For these activation result of Theorem 1.

$$\mathbb{E}\left[V(H_{\ell+1})\right] = \mathcal{O}\left((1-\alpha)^\ell + \frac{1}{\sqrt{d}}\right)$$

holds under $\mathcal{A}_1(\alpha, \ell)$. This also aligns with our conjecture for all activations as we observed that $V = Q$ for these activations.

Appendix G. Experiments for convolutional networks

The introduced initialization scheme in Section 5 extends to convnets. In convolutional networks, hidden representation are in the tensor form $H_\ell \in \mathbb{R}^{d\times m\times m\times n}$ where d is the number of filters and k is the image dimensions. Let matrix $W_\ell^{d\times k^2}$ are weights of convolutions layer ℓ with kernel size k and d filters. A 2D convolution is a matrix multiplication combined by projections as:

$$\text{conv2d}(W_\ell, H_\ell) = \text{fold}\left(W_\ell \underbrace{\text{unfold}(H_\ell)}_{H'_\ell}\right) \tag{21}$$